

Business Problem Statement

A retail company wants to understand customer shopping behavior to improve revenue, customer engagement, loyalty, and marketing effectiveness. The business observed differences in purchasing trends across demographics, categories, discounts, shipping preferences, and subscription behavior. The company requires data-driven insights to answer the following question: **How can customer shopping data be leveraged to improve customer engagement, optimize product strategy, and increase revenue?**

Data Preparation & Cleaning (Python)

- Loaded customer transaction dataset using Pandas - Checked missing values and duplicates - Standardized column names - Created age groups and transformed features - Prepared clean dataset for SQL and Power BI

Business Analysis using SQL

SQL queries were written to solve business problems such as: - Revenue by gender - Subscription impact on sales - Top-rated products - Discount-dependent products - Customer segmentation - Revenue by age group

Dashboard & Visualization (Power BI)

An interactive dashboard was built to visualize: - Revenue by category - Sales by category - Revenue by age group - Subscription distribution - KPI cards for customers, purchase amount, and ratings

Expected Business Impact

- Better customer targeting - Improved loyalty programs - Better product positioning - Revenue optimization - Smarter marketing campaigns